

Vaccinations



BY

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Vaccinations

- **Vaccination:** means the administration of antigen(s) for active immunization.
- **Vaccine:** is an antigen, originally derived from a bacterium, virus or protozoon, used for active immunization.



Types of vaccines

1. **Live attenuated organisms:** e.g. BCG
2. **Killed organisms:** e.g. Pertussis
3. **Toxoids: e.g.** tetanus toxoid
4. **DNA recombinant:** e.g. Hepatitis B
5. **Component vaccines:** e.g. Meningococcal





The Cold Chain

- The "cold chain" is a system of storage and transport of vaccines at low temperature from the manufacturer to the actual vaccination site.
- The cold chain system is necessary because vaccine failure may occur due to failure to store and transport under strict temperature controls.
- Among the vaccines, polio is the most sensitive to heat, requiring storage at minus 20 C.



The Cold Chain

- Vaccines which must be stored in the freezer compartment are: polio and measles.

- Vaccines which must be stored in the COLD PART but never allowed to freeze are: typhoid, DPT, tetanus toxoid, DT, BCG and diluents.



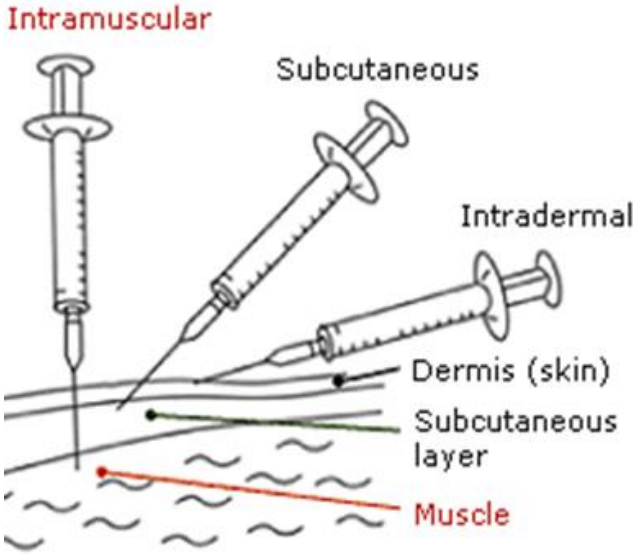


Scheme of immunization

- **One dose vaccines** (BCG)
- **Multiple dose vaccines** (polio, DPT, hepatitis B)
- **Booster vaccination: To maintain immunity level as it declines after some time has elapsed** (DT, MMR).

Routes of administration

- Subcutaneous route (MMR)
- Oral route (Sabin vaccine)
- Intradermal route (BCG vaccine)
- Intramuscular route (most vaccines)









Obligatory vaccinations in Egypt

Age	Vaccine	Route	Dose
Up to 1 week	Polio	Oral	2-3 drops
	Hepatitis B	IM	0.5 mL
Up to 3 months	BCG	ID	0.05 to 0.1 mL
2 months	Polio	Oral	2-3 drops
	DPT	IM	0.5 mL
	Hepatitis B	IM	0.5 mL
	Hib	IM	0.5 mL
4 months	Same as 2 months		
6 months	Same as 2 months		
9 months	Polio	Oral	2-3 drops
12 months	Polio	Oral	2-3 drops
	MMR	SC	0.5 mL
18 months	Polio	Oral	2-3 drops
	DPT	IM	0.5 mL
	MMR	SC	0.5 mL
	Hib	IM	0.5 mL

Obligatory vaccinations in Egypt

- **ID = intradermal, IM = intramuscular, SC = subcutaneous**
- **BCG = Bacille Calmette-Guerin**
- **DPT = diphtheria, pertussis and tetanus toxoid**
- **Hib = haemophilus influenza type b**
- **MMR = Measles – Mumps – Rubella**
- **NB. Low grade fever, mild respiratory infection, diarrhea and minor illnesses are not contraindications for vaccination.**
- **NB. Vitamin A capsule is given at 9 months of age.**

BCG vaccine



BCG Papule



BCG Ulcer

- Live attenuated vaccine
- Available in freeze-dried powder form that can be stored up to 1 year at 2-10 degree, protected from light.
- Must be given shortly after reconstitution because it deteriorates within few hours.
- Given in the first 3 months of life.
- Intradermal injection of 0.05 ml in the skin overlying the upper 1/3 of the left deltoid muscle.
- Side effects: Axillary lymphadenopathy - disseminated TB.

Poliomyelitis virus vaccines

1- Sabin vaccine (trivalent oral polio vaccine TOPV):

- Live attenuated vaccine
- Given orally
- Life long immunity
- Herd immunity
- Induce systemic and local intestinal immunity.
- Fragile and need proper storage

Poliomyelitis virus vaccines

2- Salk vaccine (monovalent):

- killed virus vaccine
- Can be used in immunocompromised individuals
- Given I.M.
- Expensive
- Easy storage
- Give only systemic immunity

DPT Vaccine

- **Triple vaccine (Diphtheria toxoid – Tetanus toxoid – Killed acellular pertussis)**
- **Tetanus toxoid is given to pregnant women to prevent tetanus neonatorum.**
- **Route: IM**

DPT Vaccine

Side effects:

1. Local reactions (generally erythema and induration with or without tenderness) are common

2. Mild systemic reactions such as fever, drowsiness, and anorexia occur frequently.

3. Neurological complications (convulsion – rarely encephalopathy)

DPT Vaccine

Contraindications for DPT vaccination:

1. Sever febrile illness

2. During polio epidemics

3. History of sever adverse reaction (convulsions – encephalopathy – anaphylaxis) and give only DT

4. Children with progressive neurologic disease

5. After 6 years of age use only DT

MMR

- Live attenuated (Measles – Mumps – Rubella)
- Available in dried form and must be stored frozen at 20 degree. After reconstitution, it should be kept in ice and used within few hours.
- Given by SC injection

❖ **Side effects:**

- Fever, malaise, rash (10 days after vaccination)
- Febrile convulsions
- Activation of dormant T.B. focus.
- Encephalitis is very rare.

Hepatitis B vaccine

- **Recombinant DNA vaccine**
- **Given by IM injection in the antro-lateral aspect of the thigh in infants and children and in the deltoid muscle for adolescents and not in the gluteal region.**
- **Given at birth and with DPT and polio vaccines at ages of 2,4 and 6 months.**
- **Minimal side effects: low grade fever and pain at the injection site.**

Haemophilus influenza type b (Hib)

- Conjugate vaccine
- Given by IM injection
- Given at 2,4,6 months and at 18 months of age




Thank You!